Equational theories

Lean 4 Formalization Companion – Chapter 6

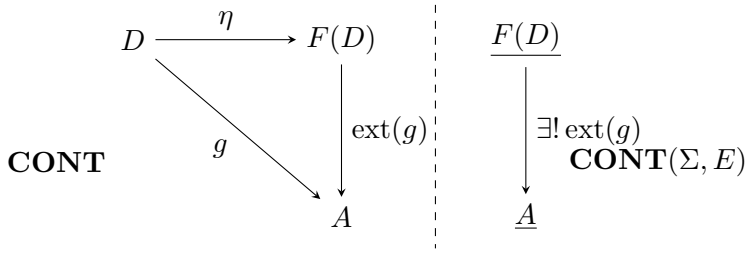
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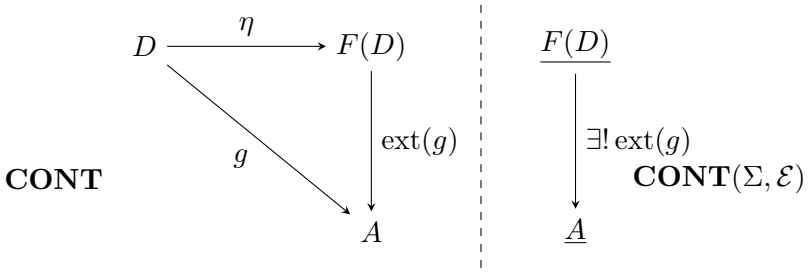
Section 6.1

Source PDF page 86

Proposition 6.1.1. Dcpo algebras form a complete category. Lean

Theorem 6.1.2. Free dcpo algebras exist for inequational theories. Lean





Source PDF page 88

Proposition 6.1.3. The syntactic basis for free continuous algebras is an abstract basis. Lean

Proposition 6.1.4. Syntactic approximation has the required inclusion properties. Lean

Source PDF page 89

Lemma 6.1.5. The quotient syntactic basis is an abstract basis. Lean

Theorem 6.1.6. Free continuous dcpo algebras are built from the syntactic basis. Lean

Source PDF page 90

Theorem 6.1.7. Free continuous-domain algebras exist. Lean

Source PDF page 91

Proposition 6.1.8. The extension map is Scott-continuous. Lean

Theorem 6.1.9. The free-algebra monad is locally continuous.

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Theorem 6.1.10. Free algebraic-domain algebras exist.

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Source PDF page 92

Proposition 6.1.11. The unit order-reflection criterion for free algebras.

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Definition 6.1.12. A strict dcpo-algebra is a dcpo-algebra whose carrier has a least element. A strict homomorphism between strict algebras is a Scott-continuous homomorphism which preserves the least element.

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Theorem 6.1.13. Free strict dcpo- and domain-algebras exist. Equivalently, the forgetful functors

$$\begin{aligned} \mathbf{DCPO}_{\perp!}(\Sigma, E) &\longrightarrow \mathbf{DCPO}_{\perp!}, \qquad \mathbf{CONT}_{\perp!}(\Sigma, E) \longrightarrow \mathbf{CONT}_{\perp!}, \\ \mathbf{ALG}_{\perp!}(\Sigma, E) &\longrightarrow \mathbf{ALG}_{\perp!} \end{aligned}$$

have left adjoints.

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Proposition 6.1.14. Finite-poset preservation yields bifinite preservation.

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Section 6.2

Definition 6.2.1. The convex, or Plotkin, powertheory has one binary operation $\cup$ and equations Lean

1. $x \cup y = y \cup x$,
2. $(x \cup y) \cup z = x \cup (y \cup z)$,
3. $x \cup x = x$.

The operation is formal union. The free dcpo-semilattice over a dcpo D is the Plotkin powerdomain $P^P(D)$.

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Definition 6.2.2. For subsets M, N of a set with relation R, the Egli-Milner relation $M \, R_{EM} \, N$ means Lean

$$(\forall a \in M \; \exists b \in N.\; aRb) \quad \text{and} \quad (\forall b \in N \; \exists a \in M.\; aRb).$$

Theorem 6.2.3. If D is continuous with basis $\langle B, \ll \rangle$, then the Plotkin powerdomain is the ideal completion of Lean

$$\langle \mathcal{P}_f(B), \ll_{EM} \rangle.$$

Source PDF page 95

Definition 6.2.4. For a subset M of a preordered set $\langle B, \sqsubseteq \rangle$, define its convex hull by Lean

$$\mathrm{Cx}(M) = \{\, a \in B \mid \exists m, n \in M.\; m \sqsubseteq a \sqsubseteq n \,\}.$$

A set equal to its convex hull is called convex.

Proposition 6.2.5. For a preordered set $\langle B, \sqsubseteq \rangle$ and subsets $M, N \subseteq B$: Lean

1. $\mathrm{Cx}(M) = {\uparrow} M \cap {\downarrow} M$,
2. $M \subseteq \mathrm{Cx}(M)$,
3. $\mathrm{Cx}(\mathrm{Cx}(M)) = \mathrm{Cx}(M)$,
4. $M \subseteq N \Rightarrow \mathrm{Cx}(M) \subseteq \mathrm{Cx}(N)$,
5. $M =_{EM} \mathrm{Cx}(M)$,
6. $M =_{EM} N \iff \mathrm{Cx}(M) = \mathrm{Cx}(N)$.

Proposition 6.2.6. If D is algebraic, then the Plotkin powerdomain of D is isomorphic to the ideal completion of Lean

$$\langle \mathcal{P}_{\mathrm{Cx},f}(K(D)), \sqsubseteq_{EM} \rangle,$$

where $\mathcal{P}_{\mathrm{Cx},f}(K(D))$ is the set of finitely generated convex subsets of $K(D)$.

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Definition 6.2.7. Hoare and Smyth powertheories. Lean

Proposition 6.2.8. Hoare and Smyth powerdomains have lattice operations. Lean

Proposition 6.2.9. Finite-basis approximations for Hoare and Smyth powerdomains. Lean

Source PDF page 97

Theorem 6.2.10. Hoare and Smyth powerdomains of continuous domains are continuous. Lean

Proposition 6.2.11. Subset approximation facts for Hoare and Smyth powerdomains. Lean

Proposition 6.2.12. Algebraic Hoare and Smyth powerdomains are algebraic. Lean

Source PDF page 98

Theorem 6.2.13. For a continuous domain D, the Hoare powerdomain is isomorphic to the lattice of all non-empty Scott-closed subsets of D. Formal union is interpreted by actual union. Lean

Theorem 6.2.14. For a continuous domain D, the Smyth powerdomain is isomorphic to $\kappa D \setminus \{\emptyset\}$, the non-empty Scott-compact saturated subsets ordered by reverse inclusion. Formal union is interpreted by intersection. Lean

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Corollary 6.2.15. Smyth powerdomains of coherent domains are bifinite-continuous. Lean

Definition 6.2.16. For a dcpo D, $\mathrm{Lens}(D)$ is the set of all non-empty subsets that are intersections of a Scott-closed set and a compact saturated set. On lenses define the topological Egli-Milner ordering by Lean

$$K \sqsubseteq_{TEM} L \quad \Longleftrightarrow \quad L \subseteq \uparrow K \text{ and } K \subseteq \mathrm{Cl}(L).$$

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Proposition 6.2.17. For a dcpo D: every lens is convex and Scott-compact; a canonical representation of a lens L is $\uparrow L \cap \mathrm{Cl}(L)$; and $\sqsubseteq_{TEM}$ is anti-symmetric on $\mathrm{Lens}(D)$. Lean

Lemma 6.2.18. Let D be continuous with basis B, and let I be an ideal in $\langle \mathcal{P}_f(B), \ll_{EM} \rangle$. Then Lean

$$\phi(I) = \{ \uparrow A \mid A \subseteq \bigcup I \text{ is directed and } A \cap M \neq \emptyset \text{ for every } M \in I \}.$$

Theorem 6.2.19. If D is an ω-continuous domain, then the Plotkin powerdomain $P^P(D)$ is isomorphic to Lean

$$\langle \mathrm{Lens}(D), \sqsubseteq_{TEM} \rangle.$$

Formal union is interpreted as union followed by topological convex closure.

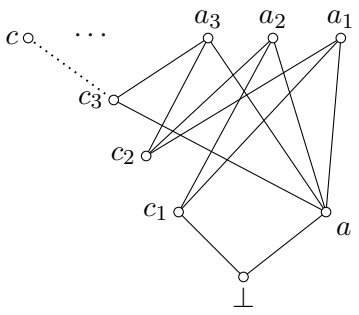


Figure 13: An algebraic domain in which topological Egli–Milner ordering and ordinary Egli–Milner ordering do not coincide.

Source PDF page 102

Lemma 6.2.20. If L is a Lawson-compact subset of a continuous domain D, then $\downarrow L$ is Scott-closed. Lean

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Corollary 6.2.21. In a coherent domain, the lenses are precisely the convex Lawson-compact subsets. For these, topological Egli-Milner ordering and ordinary Egli-Milner ordering coincide.

Theorem 6.2.22. If D is coherent, then the Plotkin powerdomain of D is isomorphic to

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$$\langle \mathrm{Lens}(D), \sqsubseteq_{EM} \rangle.$$

Formal union is interpreted as union followed by convex closure.

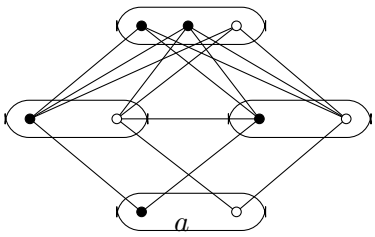


Figure 14: Part of an algebraic domain where Theorem 6.2.19 fails.